OEE-Based Manufacturing Loss Diagnostics: A Reproducible Synthetic Case Study

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***Abstract*—Overall Equipment Effectiveness (OEE) summarizes manufacturing availability, performance, and quality, but an aggregate OEE value does not by itself localize the losses responsible for poor performance. This paper presents a reproducible diagnostic workflow that combines correctly aggregated OEE, line–shift and product segmentation, Pareto loss ranking, temporal comparison, and deterministic engineering investigation prompts. The method is evaluated on a 30-day synthetic discrete-manufacturing case study comprising three production lines, three shifts, four products, and controlled loss patterns that provide known ground truth. Across 270 production records, the simulated system achieved 79.3% OEE, with 91.6% availability, 88.4% performance, and 98.0% quality. The workflow identified Line-C as the weakest line (77.0% OEE), localized the highest line–shift scrap rate to Line-B/Shift-2 (4.6%), and ranked maintenance as the largest downtime category (4,537 min). A controlled degradation interval reduced OEE from a 79.1% baseline to 73.0%, followed by recovery to 83.4%. The case study shows that the proposed workflow can recover intentionally embedded loss patterns and convert aggregate production records into prioritized investigation targets. Because the data are synthetic, the results establish reproducibility and ground-truth recovery within the simulated experiment, not industrial or causal validation.**

***Index Terms*—overall equipment effectiveness, OEE, manufacturing analytics, production losses, Pareto analysis, continuous improvement, engineering diagnostics**

I. Introduction

Manufacturing performance is commonly summarized through key performance indicators that support loss identification and improvement prioritization. Overall Equipment Effectiveness (OEE), originating in Total Productive Maintenance, combines availability, performance, and quality into a single effectiveness measure [1], [2]. OEE remains widely used, although its interpretation and implementation require explicit assumptions, especially when aggregating multiple products or production contexts [3], [4].

A limitation of a single OEE value is that it describes the magnitude of effectiveness without directly identifying the line, shift, product, time interval, or loss category responsible for poor performance. Prior research has therefore emphasized structured interpretation of OEE and the extraction of additional diagnostic information from its components and related loss data [5], [6].

This work develops and evaluates a reproducible OEE-based diagnostic framework for a synthetic manufacturing environment. Rather than evaluating only whether OEE can be calculated, the case study asks whether a workflow built around OEE can correctly recover controlled loss patterns intentionally embedded in production, downtime, and scrap records. The primary research question is:

Can a reproducible OEE-based diagnostic framework correctly identify controlled manufacturing loss patterns across production lines, shifts, products, and time periods?

The contribution is a reproducible case study that links KPI calculation to segmentation, Pareto analysis, temporal comparison, and deterministic engineering investigation prompts. The objective is not to propose a new definition of OEE, but to demonstrate a transparent workflow for moving from an aggregate effectiveness score to evidence-based investigation priorities. All experimental data are synthetic.

II. Methodology

A. Synthetic Manufacturing Dataset

The experiment represents three manufacturing lines operating three eight-hour shifts per day over 30 days. Each production record contains date, line, shift, product, planned time, downtime, ideal cycle time, total units, good units, and scrap units. Four products are modeled: Housing, Bracket, Sensor Mount, and Connector. The resulting dataset contains 270 production records.

A fixed pseudorandom seed of 42 is used for reproducibility. Separate downtime and scrap event tables preserve the same date, line, shift, and product keys as the production table. Event quantities are generated so that their sums are consistent with production-level downtime and scrap totals.

B. OEE Calculation

OEE is computed from availability A, performance P, and quality Q:

$$A = \frac{T_p - T_d}{T_p}, \tag{1}$$

$$P = \frac{\sum_i c_i n_i / 60}{T_o}, \tag{2}$$

$$Q = \frac{N_g}{N_t}, \tag{3}$$

$$OEE = A \times P \times Q, \tag{4}$$

where T_p is planned production time, T_d is downtime, T_o is operating time, c_i is ideal cycle time in seconds for product i, n_i is produced quantity, N_g is good quantity, and N_t is total quantity.

Aggregate OEE is recalculated from summed time and output quantities. Row-level OEE values are not averaged. Product-specific ideal cycle times are retained when aggregating performance. Performance values above 1.0 are capped at 1.0 so that ideal-cycle-time deviations or synthetic variability cannot produce an OEE performance contribution above 100%.

C. Controlled Ground Truth

The synthetic generator embeds moderate, discoverable patterns: Line-B/Shift-2 receives elevated scrap and quality-hold losses; Line-C receives elevated downtime, especially maintenance-related losses; Sensor Mount receives a performance penalty; days 13–17 contain a temporary degradation; and days 23–30 contain a recovery adjustment. These patterns are not supplied to the diagnostic ranking functions.

D. Diagnostic Workflow

The analytical workflow consists of five stages: (1) KPI calculation, (2) segmentation by line, shift, and product, (3) downtime and scrap Pareto analysis, (4) temporal baseline/degradation/recovery comparison, and (5) deterministic ranking of engineering findings. Recommendations are phrased as investigation targets rather than causal conclusions.

E. Ground-Truth Recovery Criteria

A controlled pattern is considered recovered when the affected segment ranks first in the diagnostic metric corresponding to the injected loss mechanism. For temporal patterns, recovery requires the degradation interval to exhibit lower OEE than the baseline and the recovery interval to exhibit higher OEE than the degradation interval. These criteria are defined before interpreting the reported results.

III. Results

A. Overall Manufacturing Performance

Across the full 30-day dataset, OEE was 79.30%. Availability was 91.58%, performance was 88.37%, and quality was 97.98%. Total production was 138,602 units, of which 135,801 were good units. Total downtime was 10,910 min and the overall scrap rate was 2.02%.

TABLE I
Overall and Line-Level Performance

Scope	OEE	A	P	Q
Overall	79.3%	91.6%	88.4%	98.0%
Line-A	81.3%	93.6%	88.3%	98.4%
Line-B	79.6%	92.5%	88.4%	97.3%
Line-C	77.0%	88.7%	88.4%	98.2%

Line-C was the weakest production line at 77.0% OEE, driven primarily by availability of 88.7%. Line-A achieved the highest line-level OEE at 81.3%.

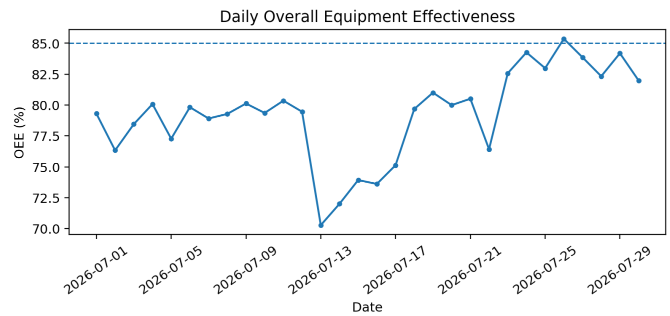

Fig. 1. Daily OEE across the 30-day synthetic experiment. The dashed reference represents the illustrative 85% target configured for this synthetic case study.

B. Line and Shift Segmentation

The lowest line–shift OEE was Line-C/Shift-3 at 75.6%. Line-B/Shift-2, although not the minimum OEE combination, exhibited the highest scrap rate at 4.6%, consistent with the embedded quality-loss pattern.

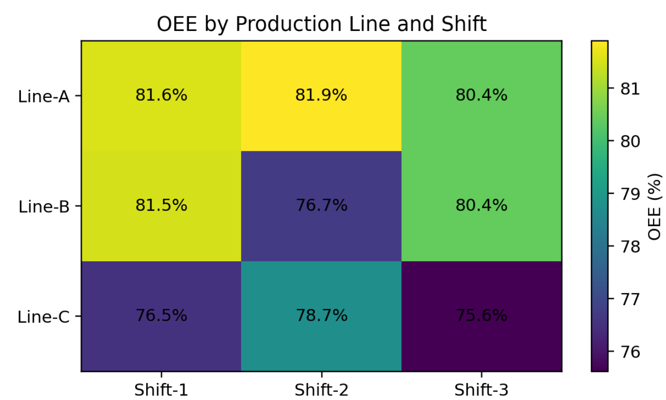

Fig. 2. OEE matrix by production line and shift.

C. Downtime and Scrap Losses

Maintenance was the largest downtime category with 4,537 min, followed by Quality hold with 1,897 min. Together they represented 59.0% of all downtime. The largest scrap category was Dimensional out-of-spec with 766 units.

D. Product Performance

Sensor Mount produced the lowest product-level OEE at 76.3%, compared with 81.2% for Housing. This result is consistent with the controlled cycle-performance penalty assigned to Sensor Mount.

E. Temporal Degradation and Recovery

The baseline period (July 1–12) achieved 79.1% OEE. During the controlled degradation period (July 13–17), OEE decreased to 73.0%, a change of -6.1 percentage points. During the recovery period (July 23–30), OEE increased to 83.4%.

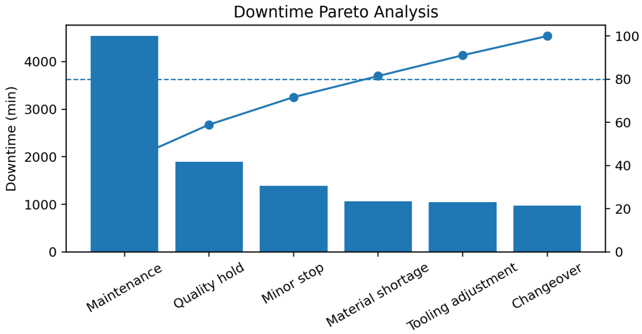

Fig. 3. Downtime Pareto analysis with cumulative contribution and 80% reference.

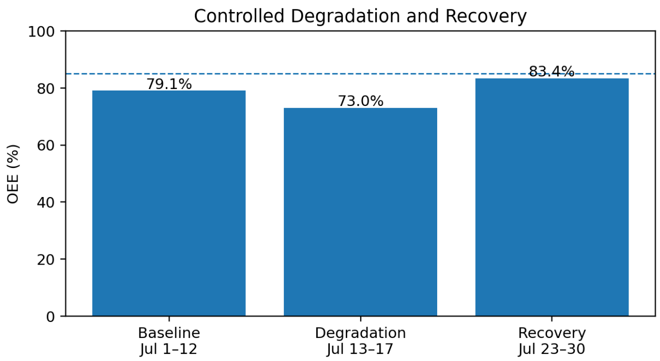

Fig. 4. OEE during baseline, controlled degradation, and recovery intervals.

IV. Engineering Diagnostic Evaluation

Table II compares the known synthetic ground truth with the observations recovered by the analytical workflow. All principal embedded patterns were visible in the corresponding diagnostic dimension.

TABLE II
Ground-Truth Recovery

Embedded pattern	Recovered evidence
Line-B/Shift-2 quality loss	Highest line–shift scrap rate (4.6%).
Line-C downtime pressure	Lowest line OEE (77.0%) and 4,902 min downtime.
Maintenance concentration	Maintenance ranked first at 4,537 min.
Sensor Mount performance pressure	Lowest product OEE (76.3%).
Temporary degradation	OEE decreased from 79.1% to 73.0%.
Recovery period	OEE increased to 83.4%.

The analysis demonstrates an important distinction between *loss localization* and *root-cause confirmation*. For example, a high scrap rate associated with a line–shift combination is sufficient to prioritize investigation, but it does not establish whether tooling, material, process parameters, operator practices, or measurement error caused the loss. The framework therefore generates suggested investigations rather than unsupported causal statements.

V. Discussion

Within this controlled experiment, the results support using OEE as an entry point for structured manufacturing diagnosis rather than as a standalone score. The overall OEE of 79.3% obscured substantially different mechanisms: Line-C was dominated by availability pressure, Line-B/Shift-2 by quality loss, and Sensor Mount by performance pressure. Segmenting the OEE components and linking them to event-level Pareto data made these differences visible.

Correct aggregation is particularly important in the presence of multiple products. Averaging row-level OEE or performance values can distort results when ideal cycle times differ. The implemented method instead aggregates ideal production minutes before calculating performance.

The synthetic ground-truth design provides a useful software-verification and analytical-evaluation mechanism. Because the expected patterns are known in advance, changes to the calculation or diagnostic code can be tested against expected behavior. However, successful recovery of synthetic patterns does not demonstrate generalization to an uncontrolled industrial environment.

VI. Limitations

This study has four principal limitations. First, all observations are synthetic and no conclusions about a real manufacturing process should be inferred. Second, the model uses daily line–shift production records rather than machine-level sensor traces, microstop data, or process parameters. Third, the deterministic insight engine ranks associations and does not perform causal inference. Fourth, process capability indices such as C_p and C_{pk} are intentionally excluded because the dataset does not contain legitimate continuous measurements with engineering specification limits.

The evaluation also uses a single fixed pseudorandom realization to ensure exact reproducibility. Consequently, this study does not estimate detection robustness across stochastic realizations. Multi-seed Monte Carlo evaluation is left for future work.

Future work should evaluate the framework using real or independently generated datasets, introduce station-level traceability, quantify robustness across multiple random seeds, and apply statistical tests when validating before/after improvements.

VII. Reproducibility and Artifact Availability

The software implementation, synthetic data generator, tests, and dashboard used in this case study are publicly available at https://github.com/rgcb01/manufacturing-oee-dashboard. The reported experiment uses a 30-day horizon and pseudorandom seed 42. The repository documents the intentionally embedded synthetic patterns and includes automated tests for OEE aggregation, Pareto calculations, period comparisons, and data consistency.

VIII. Conclusion

A reproducible OEE-based manufacturing diagnostic framework was developed and evaluated using a controlled synthetic case study. The system combined correctly aggregated OEE with line–shift segmentation, product analysis, Pareto loss ranking, temporal comparison, and deterministic engineering insights. It recovered the principal intentionally embedded patterns, including Line-C downtime pressure, Line-B/Shift-2 quality loss, Sensor Mount performance pressure, and the controlled degradation/recovery sequence. The results support treating OEE as the starting point of a diagnostic workflow rather than the final analytical output. The implementation demonstrates reproducible loss localization and prioritization within the synthetic experiment; evaluation with real manufacturing data remains necessary before making claims about industrial effectiveness.

References

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